Chapter 5: Further Supplement
Bridging Auctions & Screening in Mechanism Design

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Outline

1. Motivation: A Third Pass on Mechanism Design
2. Auctions Revisited in Mechanism Design Terms
3. Screening: Another Allocation Rule
4. Unified Mechanism Design Recap
5. Mechanism Design Recap and Pointers
6. Exercises and Additional Examples
7. Conclusion

Why a Third Pass?

- We introduced **mechanism design** first via **auctions**, then again via **screening** (price discrimination).
- We might still find the concepts confusing:
  - $x(\theta)$ as probability of "winning" in auctions,
  - $q(\theta)$ as "quantity/quality" in screening.
- Now we do a **unified comparison**:
  1. Re-interpret auctions in the same language as screening;
  2. See that "IC, IR, boundary conditions" appear identically.

Goal

Build deeper intuition: auctions and screening are two manifestations of the same mechanism design approach.

Common Threads in Auction vs. Screening

- **Both** have:
  - Agents with private types θ (valuation).
  - A **resource** to be allocated: "probability of winning" in auctions, or "quantity q" in screening.
  - **Payment** function $t(\theta)$ or $p(\theta)$.
  - **IC** ensures no misreporting, **IR** ensures participation.
- **Difference** in "who is the principal / who are the agents":
  - In auctions: the item's *owner* (seller) is the principal, many bidders (agents).
  - In screening: the *seller* is the principal facing many potential buyers (still the principal is "uninformed" about buyer's type).

Auction Model: Single Item, n Bidders

- Valuation for bidder i is θ_i, i.i.d. from distribution F.
- A **mechanism** can be seen as: each bidder reports $\hat{\theta}_i$, we assign $x_i(\hat{\theta})$ = probability that i gets the item, and $t_i(\hat{\theta})$ = payment from i.
- In a **standard** deterministic auction, $x_i(\theta) = 1$ if θ_i is highest, 0 otherwise. But we can generalize to random allocations or Myerson's approach.
- The utility for type θ_i is:

$$u_i(\theta_i) = \theta_i \cdot x_i(\theta_i) - t_i(\theta_i).$$

IC & IR

No bidder i wants to misreport $\hat{\theta}_i$; if $\theta_i = 0$, presumably $u_i(0) = 0$ is the boundary.

First-Price Auction in This Framework

- We can interpret first-price sealed-bid as:
  - $x_i(\hat{\theta}) = 1$ if $\hat{\theta}_i$ is the highest, else 0,
  - $t_i(\hat{\theta}) = \hat{\theta}_i$ if i wins,
  - So if true type is θ_i, in equilibrium you choose $\hat{\theta}_i = b^*(\theta_i)$, typically $b^*(\theta_i) < \theta_i$ but solves the optimum shading formula.
- **IR** is typically not binding for higher types, but we ensure they have nonnegative utility in equilibrium.
- **IC** is the condition that you can't do better by lying about θ_i.

Second-Price Auction in This Framework

- $x_i(\hat{\theta}) = 1$ if $\hat{\theta}_i$ is highest, else 0,
- $t_i(\hat{\theta}) = \max_{j \neq i}\{\hat{\theta}_j\}$ if i is highest,
- Then $b^*(\theta_i) = \theta_i$ is a (weakly) dominant strategy.

Hence

We see how the standard auctions can be recast as direct revelation or indirect, but we can unify them: $x(\theta)$ is the "allocation," $t(\theta)$ is the "payment," subject to IC, IR, etc.

Screening: $q(\theta)$ Instead of Probability

- In screening, a single agent with type θ chooses from a menu, or in the direct version, we imagine they "report" $\hat{\theta}$.
- Then the mechanism sets $q(\hat{\theta})$ = quantity or quality, $p(\hat{\theta})$ = price/payment.
- Utility $u(\theta) = \theta \cdot q(\theta) - p(\theta)$.

Compare to Auction

- $x(\theta)$ in auctions is either "the chance you get the item." - $q(\theta)$ in screening is "the quantity you get." They are both **allocation functions** mapping type θ to "what you receive."

IC and IR in Both Contexts

Auction	**Screening**
$x(\theta) = \Pr(\text{winning} \mid \theta)$,	$q(\theta)$ = quantity or quality,
$t(\theta)$ = expected payment.	$p(\theta)$ = price paid.
$U(\theta) = \theta x(\theta) - t(\theta)$.	$U(\theta) = \theta q(\theta) - p(\theta)$.
IC: no gain from misreport	IC: no gain from misreport
IR: $U(\theta) \geq 0$, maybe $U(0) = 0$.	IR: $U(\theta) \geq 0$, maybe $U(\underline{\theta}) = 0$.

Envelope Theorem

In each case, $U'(\theta) = \text{allocation}(\theta)$, so $U(\theta) = \int x(\theta)d\theta$ or $\int q(\theta)d\theta$ if boundary is zero.

Single-Crossing Condition (Monotonic Allocation)

- Auction: If θ is higher, your probability of winning $x(\theta)$ should not be smaller. Usually $x(\theta)$ is nondecreasing in θ.
- Screening: If θ is higher, we typically serve a bigger $q(\theta)$ (monotonic in θ).
- **Single-crossing** ensures that an agent with higher θ always "prefers" a higher q or a higher chance of winning, making the design consistent with a monotonic $x(\theta)$ or $q(\theta)$.

Hence

Whether it's $x(\theta)$ or $q(\theta)$, the same envelope logic arises:
$u'(\theta) = \text{allocation}(\theta)$, bounding the shape of $u(\theta)$.

Mechanism Design in a Nutshell

- **Principal** wants to assign "something" (probability, quantity) to an agent with type θ, and possibly collect a payment $t(\theta)$.
- **Agent's utility** $= \theta \cdot$ allocation $-$ payment (or similar).
- **IC condition** ensures no type gains by lying about θ.
- **IR condition** ensures a type with θ is not made worse than opting out.

Examples

- Auction: principal is the seller, many agents are bidders. - Screening: principal is the seller, each buyer is an agent. - Even in public goods or procurement, same structure holds with roles possibly reversed or multiple goods.

Comparing Auction & Screening Solutions

- **Auction** with n bidders: typically, allocate $x_i(\theta)$ to the highest θ_i, or do Myerson's virtual value approach for revenue.
- **Screening** with α fraction of type H, or continuum distribution:
  - We saw how the principal "excludes" or "distorts" certain types if that increases profit or ensures no misreporting.
- **Both** solve an integral or summation problem to account for the envelope formula for $u(\theta)$.

When & Why They Differ

- **Auction scenario**: multiple potential buyers, we choose who "wins."
  - The monopoly power is on the item owner side.
  - If we do second-price, it's efficient. If we do Myerson's approach, might skip some buyer(s).
- **Screening scenario**: typically a single buyer at a time, or we replicate for many buyers each.
  - The principal chooses $q(\theta)$ for each buyer.
  - Over many buyers, it's multiple "monopoly screens" in parallel, not direct competition among them.

Still the same IC/IR logic under the hood.

Mathematical Unification

Auction:

$$\text{profit} = \int t(\theta)\, dF(\theta), \quad t(\theta) = \theta \cdot x(\theta) - u(\theta).$$

Screening:

$$\text{profit} = \int [p(\theta) - c(q(\theta))]\, dF(\theta),$$

with $p(\theta) = \theta q(\theta) - u(\theta)$.
In both:

$$u'(\theta) = \text{allocation}(\theta), \quad u(\underline{\theta}) = 0.$$

Hence the "rent" function $u(\theta)$ is pinned down, leaving the principal to choose allocation(θ) to maximize net payoff subject to monotonicity constraints (single-crossing).

Remark

Thus the mathematics is the same shape, just different naming.

Conclusion of the Comparison

- **Both** auctions and screening revolve around designing allocation + payment to handle private info.
- **Both** yield an "IC-based" constraint that shapes a "rent" or "information rent" function $u(\theta)$, pinned by $u(\underline{\theta}) = 0$ or $u(0) = 0$ boundary.
- They differ mainly in structure: auctions have multiple agents competing for a single resource, screening is typically one agent "buying different qualities" from a single principal. But mathematically, *it's the same mechanism design logic.*

Revelation Principle (Brief Recap)

Claim: If an outcome can be implemented by a complicated mechanism, then there is a *direct* mechanism in which each agent "reports" their type, and telling the truth is an equilibrium that achieves the same outcome.
Why we restate: - In both auctions and screening, you can do "direct revelation"—the agent says θ, we pick $x(\theta)$ or $q(\theta)$, then a payment $t(\theta)$ or $p(\theta)$.
- This consistent approach helps unify the formal solutions.

Hence

Focus on direct, truthful approach, derive $u(\theta)$ from $u'(\theta) = \text{allocation}(\theta)$.

IC, IR, Boundary Condition

- **IC** ensures the agent with real type θ does not prefer to pick an alternative type $\hat{\theta}$:

$$\theta \cdot \text{allocation}(\theta) - \text{payment}(\theta) \;\geq\; \theta \cdot \text{allocation}(\hat{\theta}) - \text{payment}(\hat{\theta}).$$

- **IR** ensures $u(\theta) \geq 0$. Usually we set $u(\underline{\theta}) = 0$ if $\underline{\theta}$ is the lowest type who can either accept or not.
- This boundary condition yields a unique "rent function" $u(\theta)$ once we fix the allocation function's shape.

Remark

In auctions: *highest type is allocated the good.* **In screening**: *a type θ is allocated $q(\theta)$.*

Envelope Theorem (Again and Again)

- $\frac{d}{d\theta} u(\theta) = \text{allocation}(\theta)$.
- So $u(\theta) = \int_{\underline{\theta}}^{\theta} \text{allocation}(t)\, dt$, if $u(\underline{\theta}) = 0$.
- The principal's "profit" = integral of $(\theta \cdot \text{allocation}(\theta) - u(\theta))$ minus costs, or a variant of that.
- This yields an integral expression we maximize w.r.t. the shape of allocation(θ), subject to allocation(θ) being nondecreasing (single-crossing).

Hence the same formula applies to auctions (x) or screening (q).

Example: A Simple Auction vs. Screening Parallel

Auction:

- 2 bidders, each $\theta_i \in [0, 1]$. Probability of winning for type θ_i is $x(\theta_i) = \ldots$ (something we define).
- $u_i(\theta_i) = \theta_i x(\theta_i) - t_i(\theta_i)$.

Screening:

- 1 buyer, type $\theta \in [0, 1]$. The firm picks $q(\theta)$, charges $p(\theta)$.
- $u(\theta) = \theta\, q(\theta) - p(\theta)$.

In both: IR $\Rightarrow u(0) = 0$. IC $\Rightarrow u'(\theta) = \text{allocation}(\theta)$. We can write out the integral form. The difference is just how we interpret that "allocation function."

Additional Examples to Try

- **3-type Auction** vs. **3-type Screening**:
  - Solve them side by side, see the same structure: all constraints (IC1, IC2, IRlowest, etc.) plus objective function. The final solutions appear quite parallel.
- **Multi-product screening** or **multi-item auctions**:
  - Mechanism design can handle synergy or correlation, but it gets more complicated.
  - The same "rent extraction vs. distortion" trade-off remains.

Remark

Seeing them side-by-side cements the notion that mechanism design is a broad unifying approach.

Exercises (1)

1. **Compare Auction vs. Screening Envelope**:
   - Write $x(\theta) =$ prob of winning in an auction, $q(\theta) =$ quantity in screening. Show in both we get $u'(\theta) = x(\theta)$ or $u'(\theta) = q(\theta)$.
   - Argue how $u(\theta_{\min}) = 0$ sets the boundary.
2. **2-type Auction vs. 2-type Screening**:
   - Auction: Let $n = 2, \alpha$ fraction with high $\overline{\theta}$, etc. Solve for first-price or second-price.
   - Screening: Let α fraction high $\overline{\theta}$, $(1-\alpha)$ fraction low $\underline{\theta}$. Solve for $(p_H, q_H), (p_L, q_L)$.
   - Compare the constraints' structure.

Exercises (2)

1. **Single-Crossing Condition**:
   - In an auction context, show that if type θ is bigger, it strictly prefers higher chance of winning (assuming linear payoff).
   - In screening, explain why if type θ is bigger, it strictly prefers bigger $q(\theta)$ under single-crossing condition.
   - Conclude monotonic allocation(θ) is natural.

Further Thoughts

- We introduced **Myerson's Auction** as the general revenue approach in multi-bidder auctions.
- Screening extends similarly if multiple buyer segments exist, or if we interpret a continuum distribution. The solution parallels Myerson but typically "per-buyer."
- Real-life markets can have cross-externalities, synergy, repeated interactions, etc. Mechanism design grows quite intricate.

Nevertheless

The core principle—IC, IR, boundary conditions, and the envelope theorem—remains the same across auctions or screening.

Concluding Remarks (1/2)

- **Auction**: "winning probability $x(\theta)$" if you have the highest "bid" or "virtual value." Payment $t(\theta) = \theta x(\theta) - u(\theta)$.
- **Screening**: "quantity $q(\theta)$," payment $p(\theta) = \theta q(\theta) - u(\theta)$.
- Both are single-dimensional design problems: we want to pick an allocation function that *optimizes* some objective (efficiency, revenue) while respecting **IC, IR**.

Hence

They are essentially the **same** shape of problem, just interpreted differently: who is the principal, who are the agents, how is the resource allocated.

Concluding Remarks (2/2)

- We see mechanism design thrice:
  1. **Auction** lens,
  2. **Screening** lens,
  3. **Unified** bridging lens.
- We hope this repeated exposure clarifies the **core mathematics** and logic of *IC* + *IR* + *boundary* leading to *envelope solutions*.
- Real markets are messy, but the essence is in these building blocks.

Questions?

Thank you for your attention!